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From Micro to Macro: Unsupervised FE2 Acceleration using ddfenicsx and micmacsfenicsx

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The FE2 approach [4] consists of solving a multilevel nonlinear finite element problem and is typically applied to problems with clear scale separation, where microscopic heterogeneities are much smaller than the structural scale. Resolving these heterogeneities directly at the structural level would require prohibitively fine meshes. Computational homogenisation provides consistent coupling between scales [5] (e.g. coarse-scale stress is the average of the microscale stress field). However, FE2 remains computationally expensive and memory intensive, which limits its practical use. This tutorial presents the synergic use of two libraries: i) ddfenicsx [1], which integrates Data-Driven Computational Mechanics (DDCM) [6] while preserving a familiar UFL-based workflow (see the ddfenicsx presentation abstract for details); and ii) micmacsfenics [2], an FE2 implementation featuring (i) nested FEM solvers for local problems at each coarse-scale Gauss point, (ii) efficient computation of consistent homogenised tangents (no finite differences), and (iii) support for common non-local microscale kinematical constraints (e.g. periodic, zero-average, etc). The proposed acceleration relies on a recent approach [3] that constructs strain-stress datasets on-the-fly through computational homogenisation. The method is fully unsupervised: no offline database generation or neural-network training is required [7]. Although the method focuses on solid mechanics, the methodology and libraries are general and can be easily extended to other applications. Attendees will run: i) a single-scale hyperelastic problem; ii) a heterogeneous microscale model using micmacsfenics; iii) a reference FE2 solution; and iv) a DDCM problem with a self-evolving dataset generated by the microscale solver using ddfenicsx. All scripts will be provided in a GitHub repository. Requirements are FEniCSx 0.10, dolfin_mpc, and scikit-learn.

References

[1] Rocha, F. (2023-). ddfenics: A FEniCS(x)-based (model-free) data-driven computational mechanics implementation [Software]. Zenodo. DOI: 10.5281/zenodo.7646226

[2] Rocha, F. (2023-). micmacsfenics: A FEniCs-based implementation of two-level finite element simulations (FE2) using computational homogenization [Software]. Zenodo. DOI: 10.5281/zenodo.7643777

[3] Rocha, F., Platzer, A., Leygue, A., & Stainier, L. (2025). On-the-fly adaptive sampling strategy for data-driven computational mechanics: Applications to computational homogenisation. Mechanics of Materials, 207, 105382. DOI: 10.1016/j.mechmat.2025.105382

[4] Feyel, F. (2003). A multilevel finite element method (FE2) to describe the response of highly non-linear structures using generalized continua. Computer Methods in Applied Mechanics and Engineering, 192(28-30), 3233-3244. DOI: 10.1016/S0045-7825(03)00348-7

[5] de Souza Neto, E. A., Blanco, P. J., Sánchez, P. J., & Feijóo, R. A. (2015). An RVE-based multiscale theory of solids with micro-scale inertia and body force effects. Mechanics of Materials, 80(A), 136-144. DOI: 10.1016/j.mechmat.2014.10.007

[6] Kirchdoerfer, T., & Ortiz, M. (2016). Data-driven computational mechanics. Computer Methods in Applied Mechanics and Engineering, 304, 81-101. DOI: 10.1016/j.cma.2016.02.001

[7] Zlatic, M., Rocha, F., Stainier, L., & Canadija, M. (2024). Data-driven methods for computational mechanics: A fair comparison between neural networks based and model-free approaches. Computer Methods in Applied Mechanics and Engineering, 431, 117289. DOI: 10.1016/j.cma.2024.117289